

Spark Repair Estimator

1. Most Interesting Design / UX Decision

The biggest UX choice was to shift the running total and progress into a persistent bottom ledger bar. The user isn't sitting at a desk, rather walking through a distressed property, with one thumb, lack of light and no signal. The only number that could not be lost in the thumb zone is the ledger bar, whilst the item rows remain dense: full item name, price/unit, quantity, line total. I also made room management explicit by putting an Edit control rather than building destructive actions in the gestures.

2. What Is Broken or Fragile

The most vulnerable area is install behavior as mobile browsers do not behave the same with PWA. Android Chrome will show a native install prompt, and the iOS Safari browser will ask to add to the home screen, so the app offers a fallback install prompt. Furthermore, OCR is fragile: it runs offline and does not create serials itself, but the accuracy is dependent on the photo quality, the presence of glare, focus and whether the Tesseract assets were cached since loading. localStorage is the quota that limits photo storage, images are scaled down to 1280px JPEGs, and if the quota is still violated, the app stores the estimated data, informing the user instead of silently failing.

3. Creative Addition

The creative add-on is the Offline Field Assistant and review engine: a chat over the live estimate that is deterministic, not an LLM, but is a scored keyword-intent engine that reads only the project state and the embedded price list. It marks probable missed costs before export, creates a scope summary that can be copied and pasted to other contracts, adds materials by unit, provides one-turn follow-up answers to questions about prices and progress ("and the kitchen?"), allows for defining of field terms from a built-in glossary ("what does plank mean?") and adds chips to each answer that deep link into the specific room and group. I selected this one because the worst thing that can be done in a big field is slow entry, but rather forgetting to buy a scope that will cost them money before they even reach the decision to buy.

4. What I Would Ship Next With Two More Days

If I had another day, I would have made the conflict handling stronger with more two tabs, an export progress view for super large photo packages, and a first run sample walkthrough. I'd also work on making photo tagging a more streamlined process to tie a serial plate and damage shots to room, group and item with fewer taps.

AI Tool Role

AI tools were used as a development assistant for implementation, debugging, UI polish, acceptance-test planning, and regression checks. Final product decisions, manual device testing, and submission review were performed by me.